

Session 03 : Objective

pdf/csul/text etc
python → langchain

Pluton

→ Langchain

→ (*)

□ Document loader and chunking

Document Retrieval

Similarity Search

Similarity Search with threshold

MMR

DM 25

maximum marginal Relevance

TFidf

☒ Vector database

SQL db

VS

vector database

Data structure

Time and space complexity

ANN

ENN

- indexing

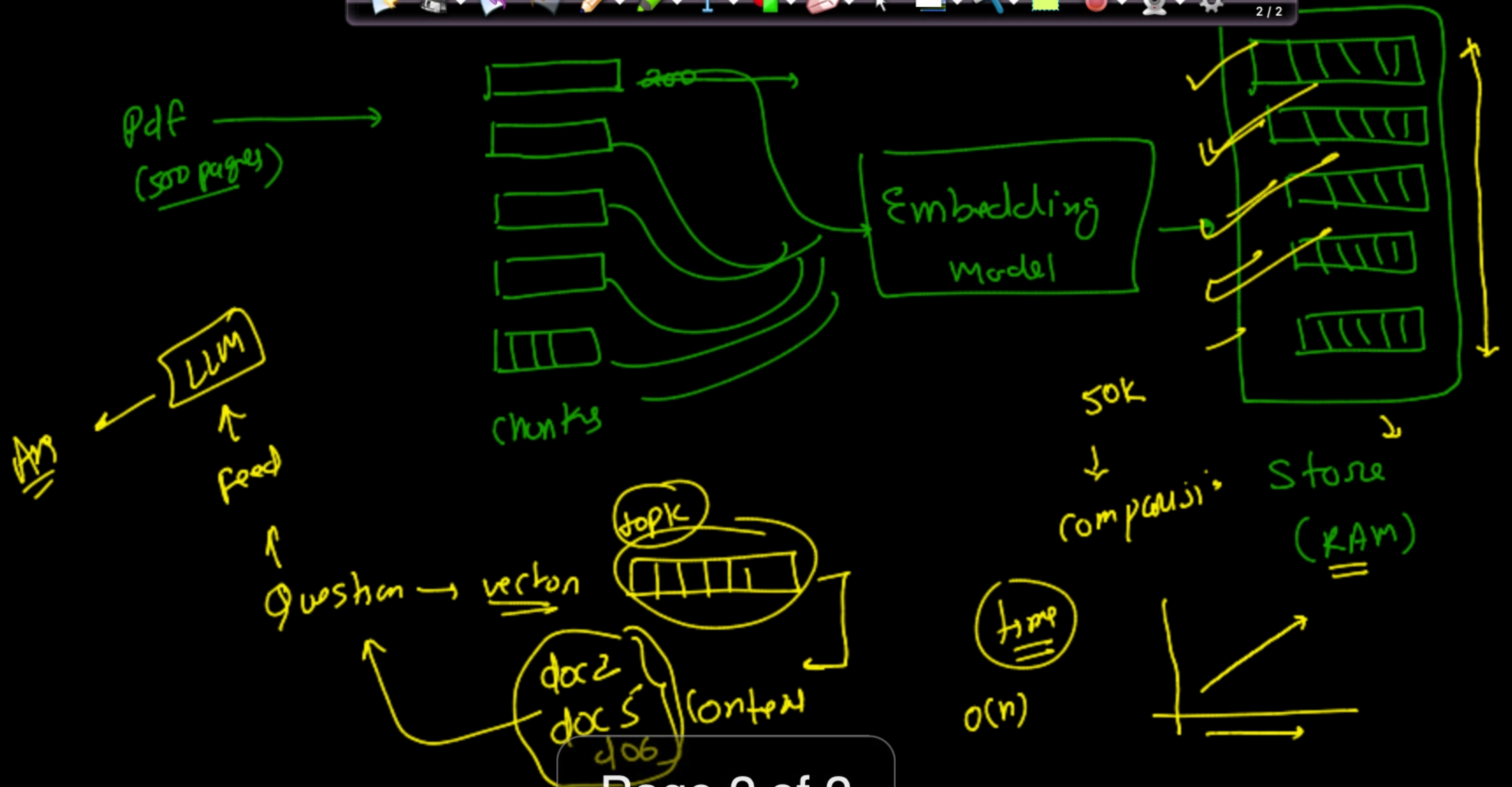
HNSW

~~TYPE~~

Goal

BFL Bot

upcoming session



XBCG

text
↓
similarity

similarity

'keyword' & BM25(keyword)

RRF

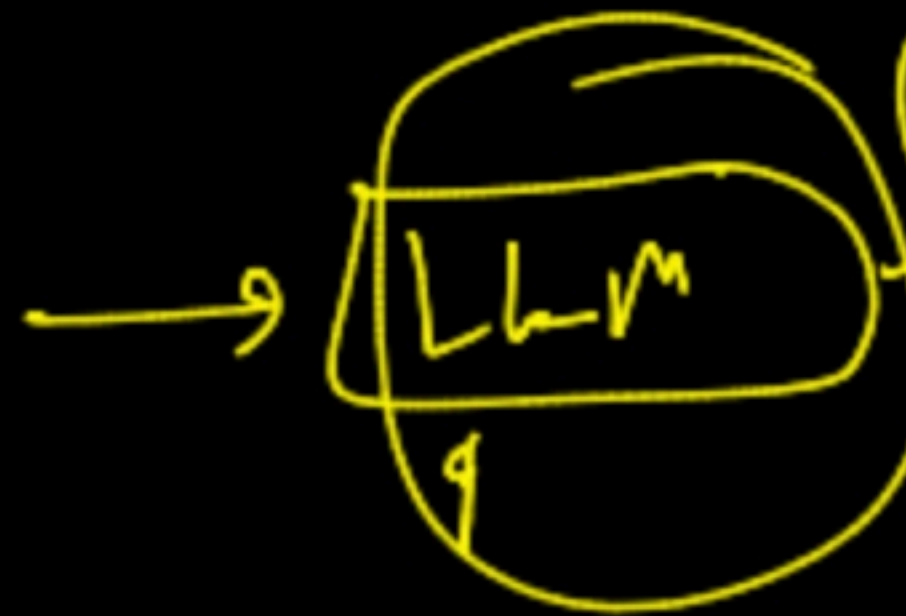


LLM

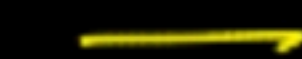


Multi-Query Retrieval

what is interest for PL



Ret()



MPR

S-6

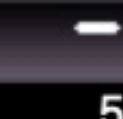
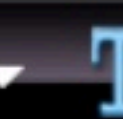


chunking

splitting

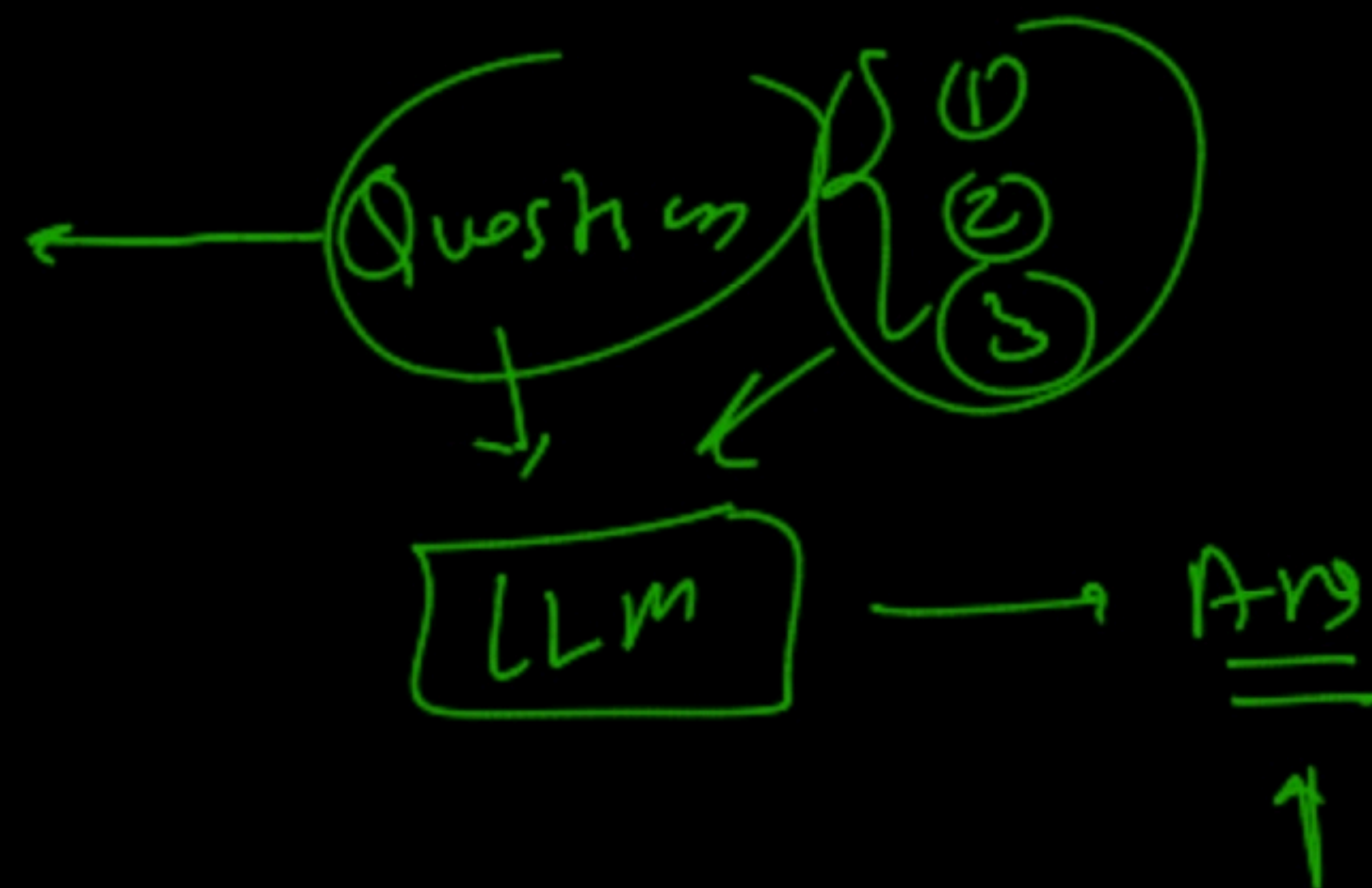
dev

production



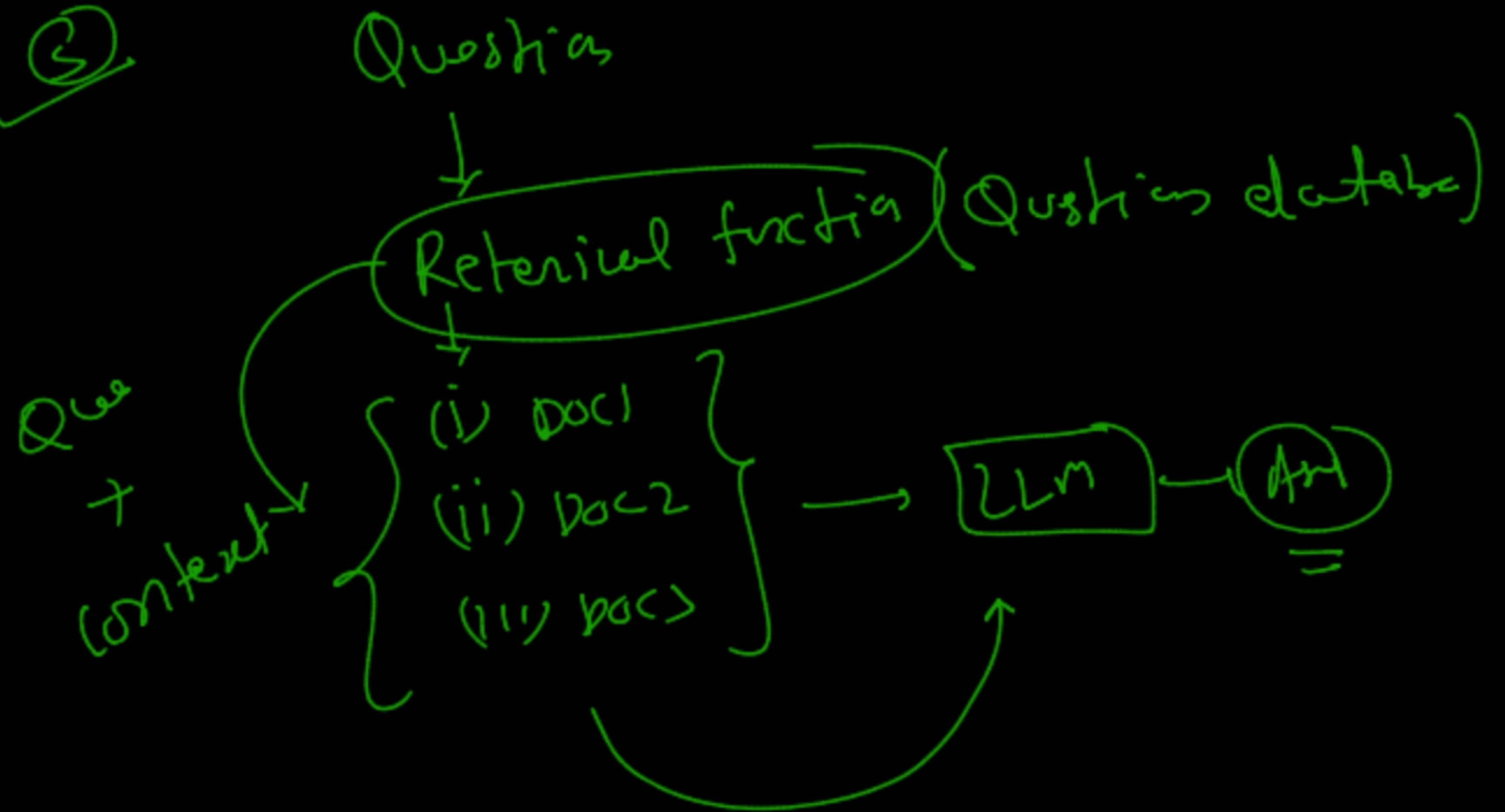
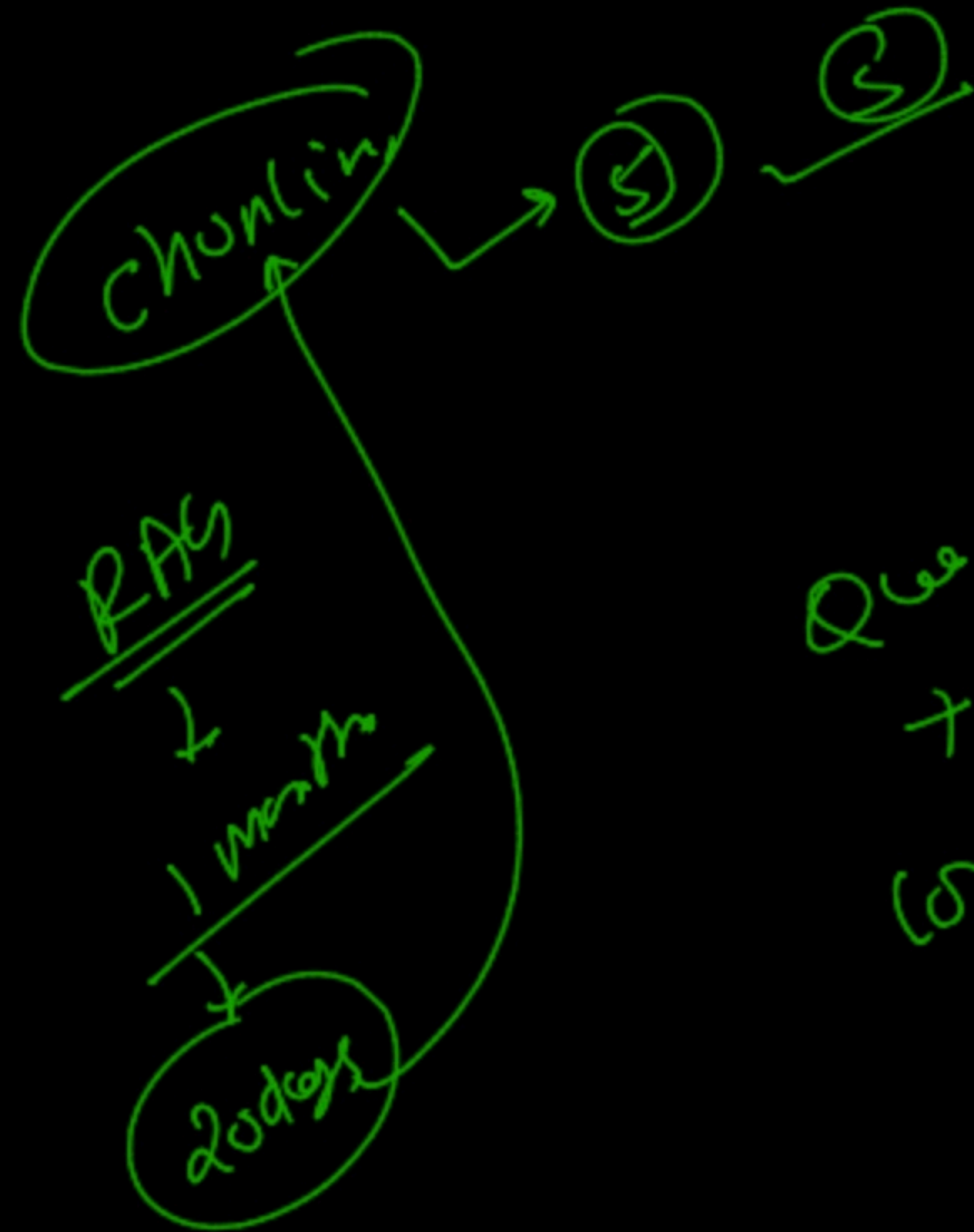


develop



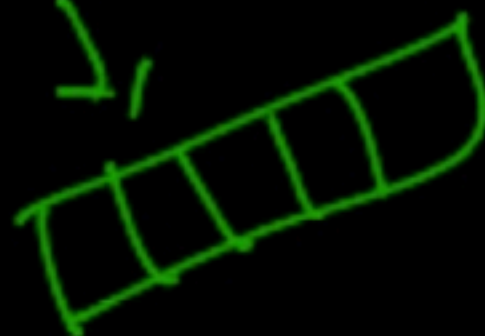
production





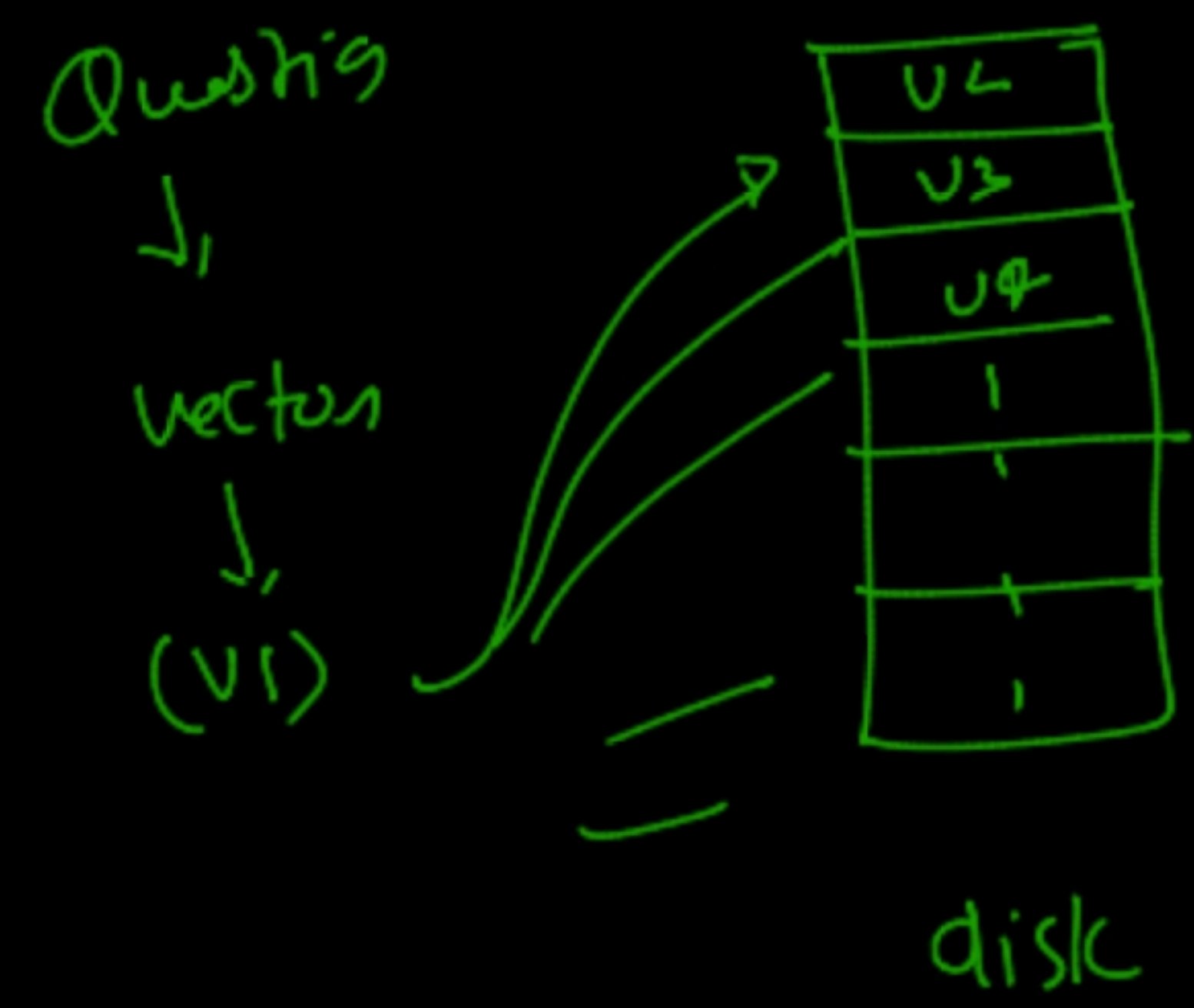
amazon
↓

t-shirts
↓



category = "PL Policy"

Question
↓
category → PL Policy
↓
meta → SQL



3000 chunks $\rightarrow$ 3000 vectors

Questions $\rightarrow$ LLM

content()

(Ques, VI) $\rightarrow$ 0.66

3000

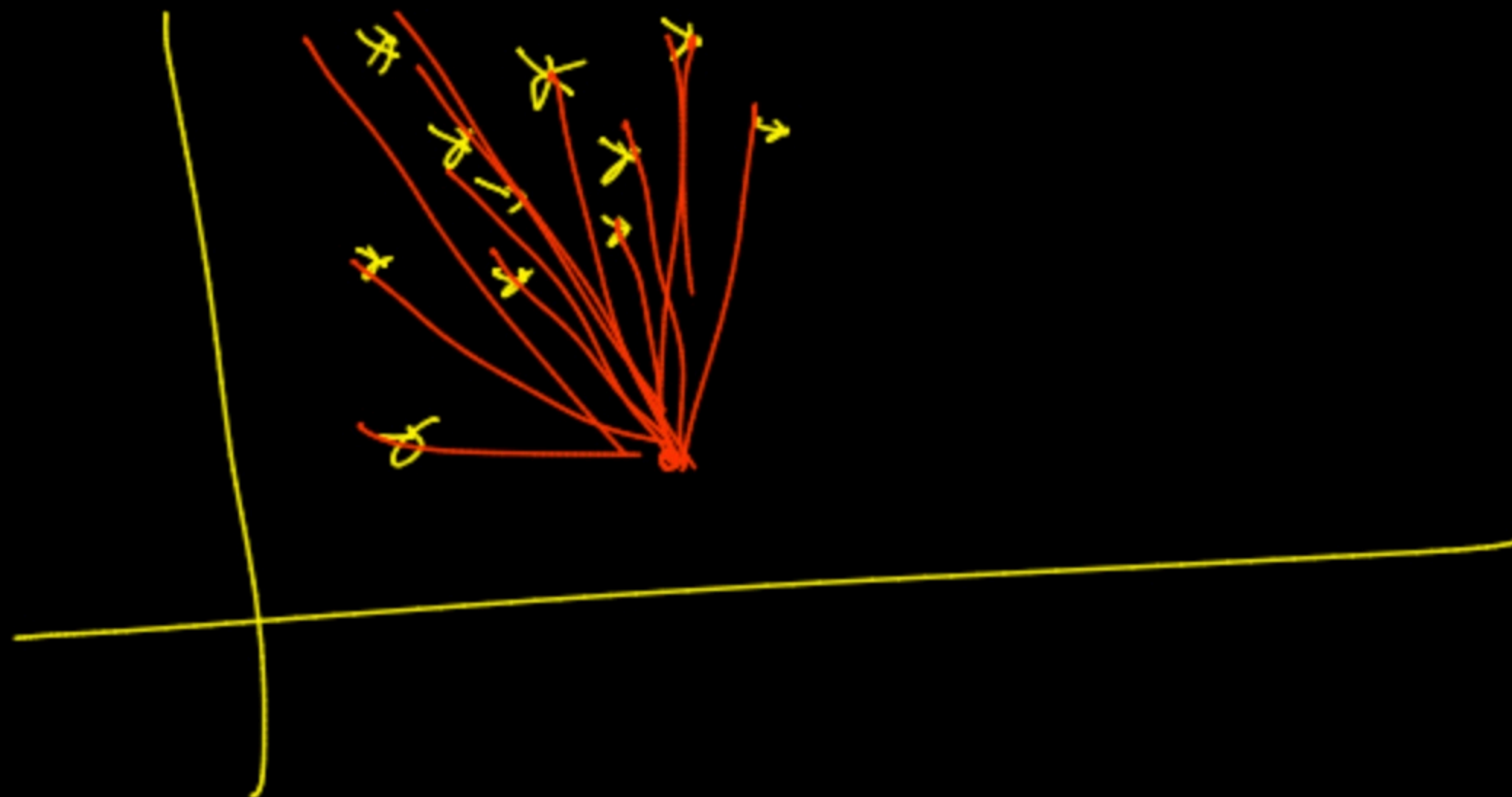
Time

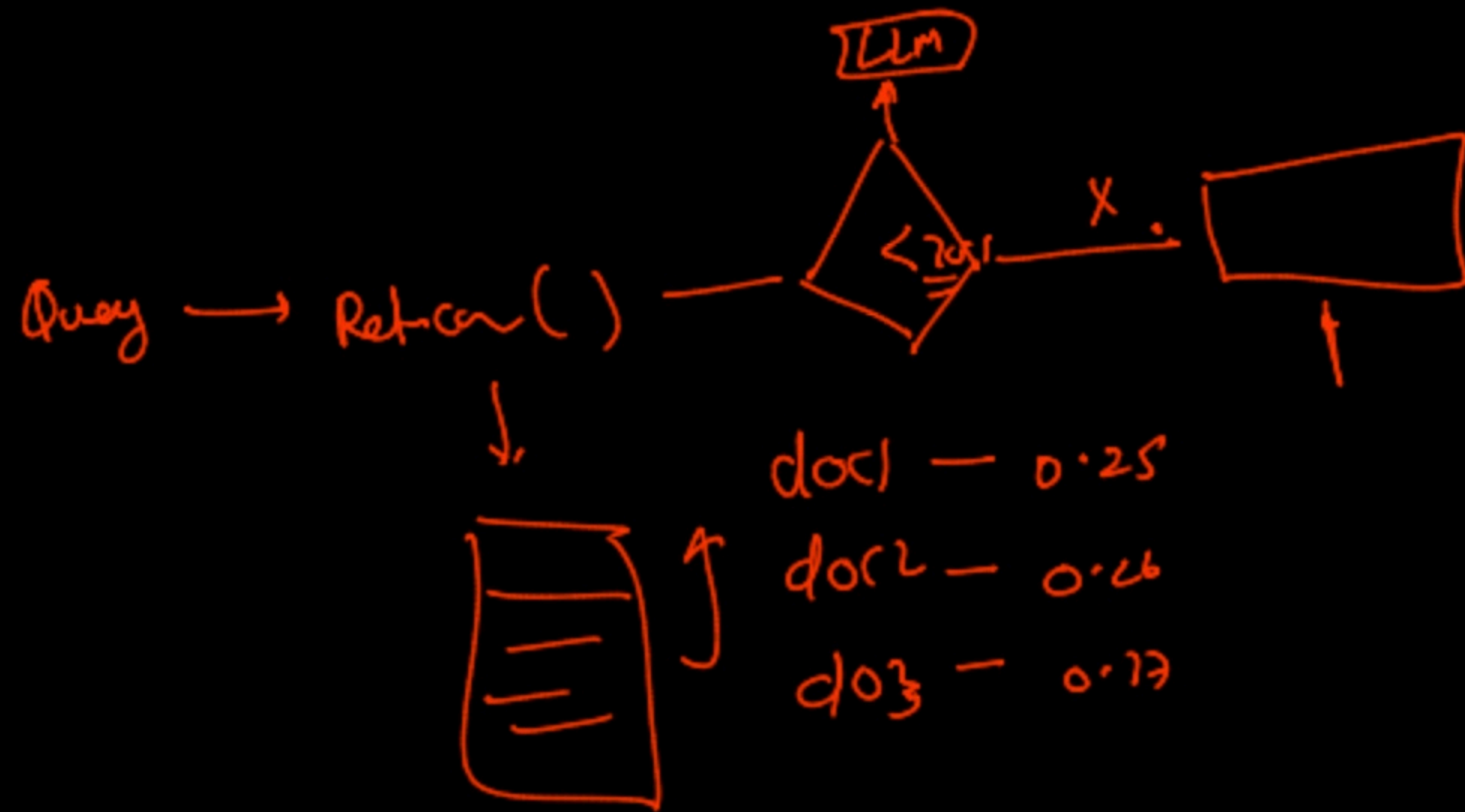
END



3Millions
↓

n times
+
3millis
+
EANN
→ ANN





Question → Retorial (Queshs)

↓
python frnctins
||

Prompt

Lkm

* * *
* * *
* * *

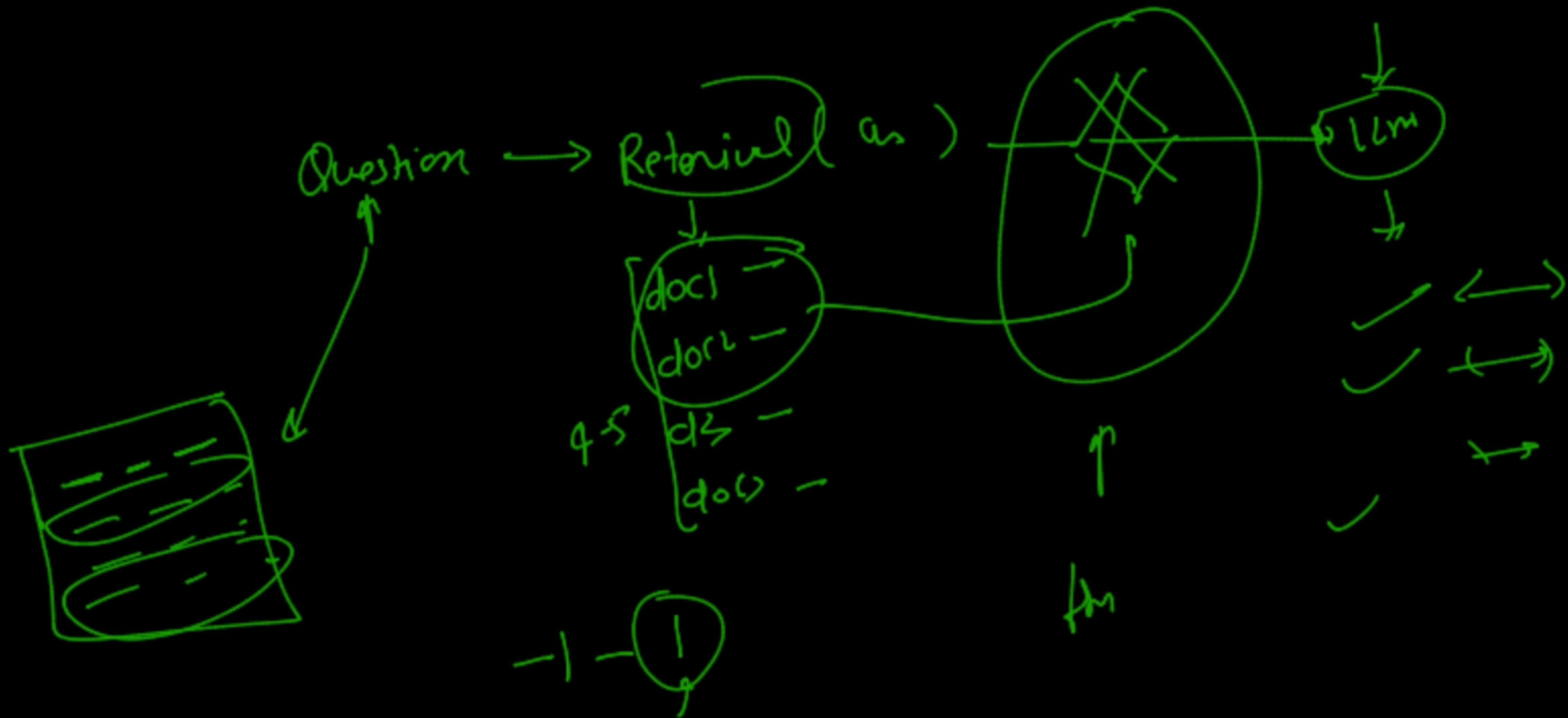
(query, embeddin)

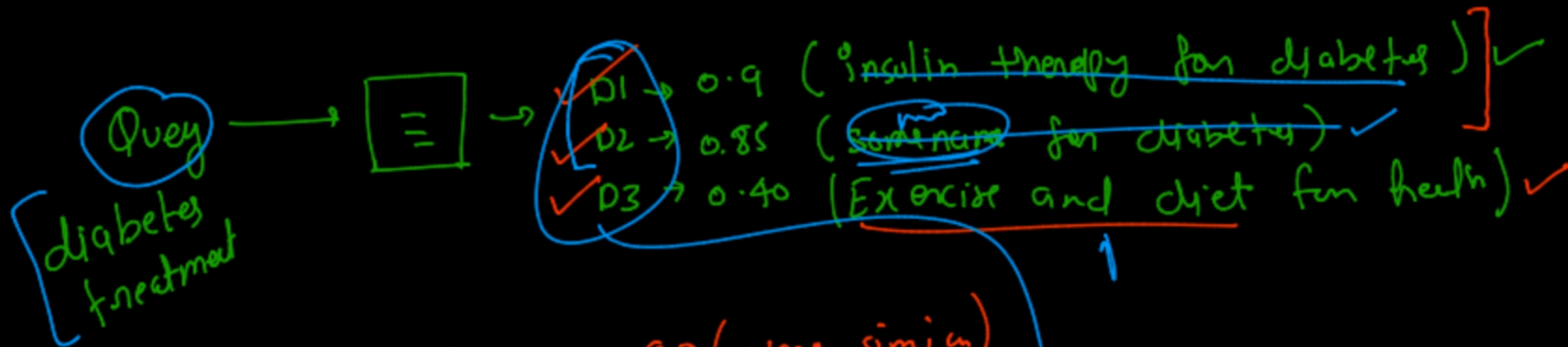
top 3

docu

0.1

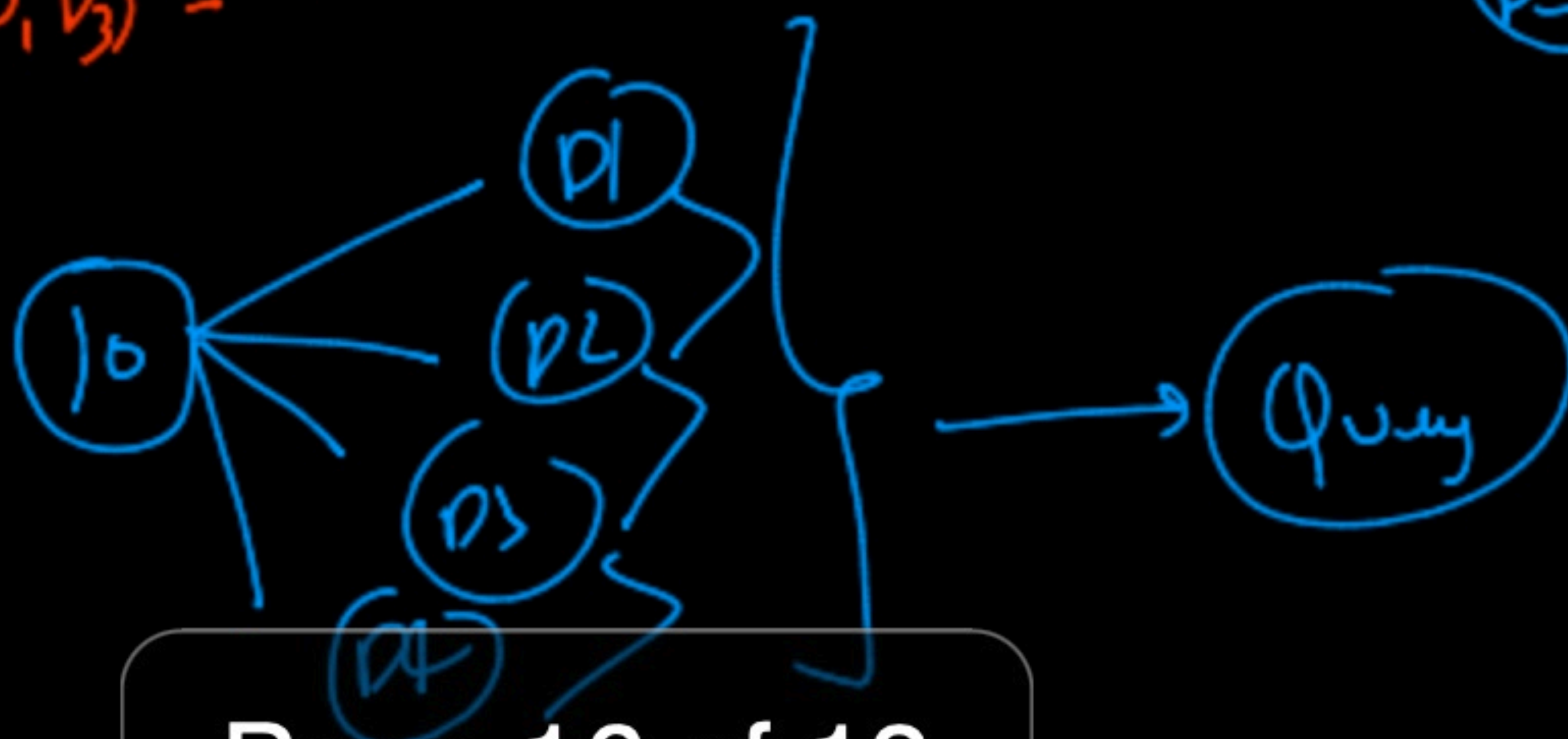
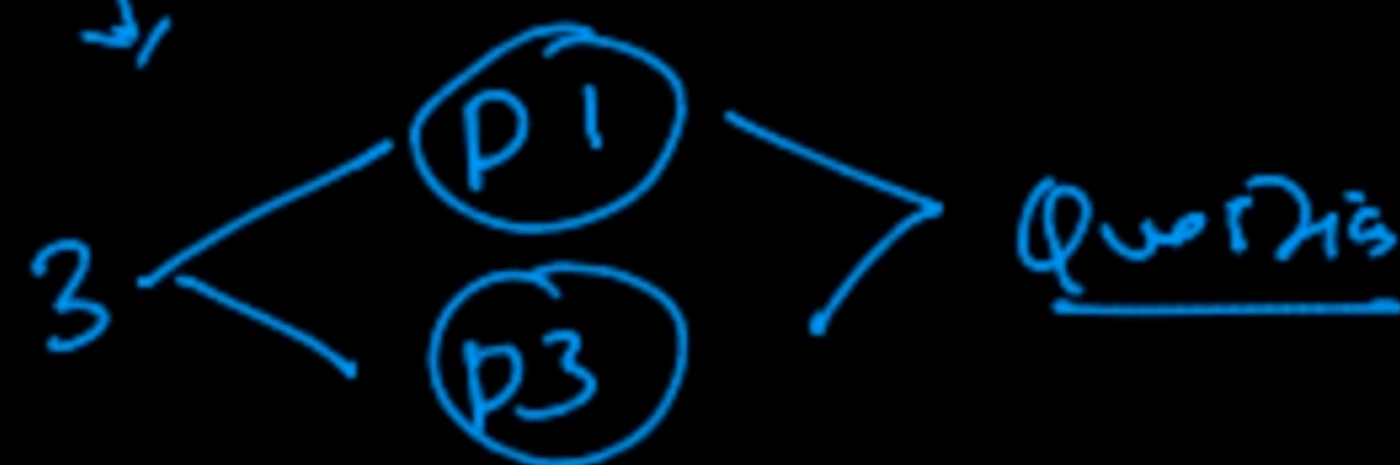






$$\text{sim}(\underline{D_1}, \underline{D_2}) = 0.88 \text{ (very similar)}$$

$$\text{and } \text{sim}(D_1, D_3) = 0.20$$



$$MMR = \underset{d \in R \setminus S}{\operatorname{argmax}} \left[\lambda \cdot \operatorname{sim}(d, q_i) - (1 - \lambda) \max(d_i, d_j) \right]$$

q : Query

R : All candidate

S : Selected docum

λ : Balance knob

($\lambda = 1 \Rightarrow$ pure relevance)
 $\lambda = 0 \Rightarrow$ pure diverse)

Query $\rightarrow \cos(v)$

$\hookrightarrow \begin{matrix} p1 \\ p2 \\ p3 \end{matrix} \} R \rightarrow \textcircled{2}$

$\begin{matrix} 1 \\ 3 \\ \wedge \\ \textcircled{p1 \ p3} \end{matrix}$





$$\lambda = 0.5$$

$$\begin{aligned} \underline{\text{MMR}}_{(p, p_2)} &= 0.5 * \underline{\text{sim}(p, q)} * \max(\text{sim}(p, p_2)) \\ &= 0.5 * 0.85 - \max(0.5, 0.6) \end{aligned}$$

$$\text{MMR} = \cancel{0.45} - 0.015$$

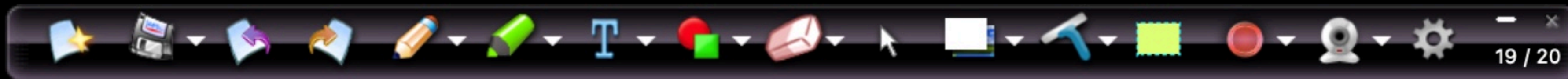
$$\text{MMR} = 0.5 * \text{sim}(p_3, q) - \max(\text{sim}(p_3, p_1))$$

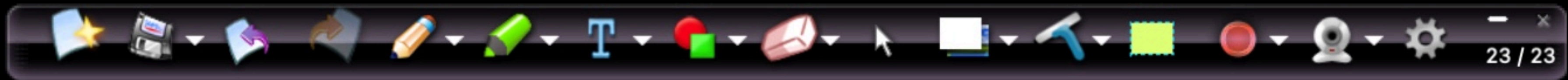
$(p_3) \rightarrow$

$$\boxed{\text{MMR} = \underline{\underline{0.10}}}$$

(p_3)

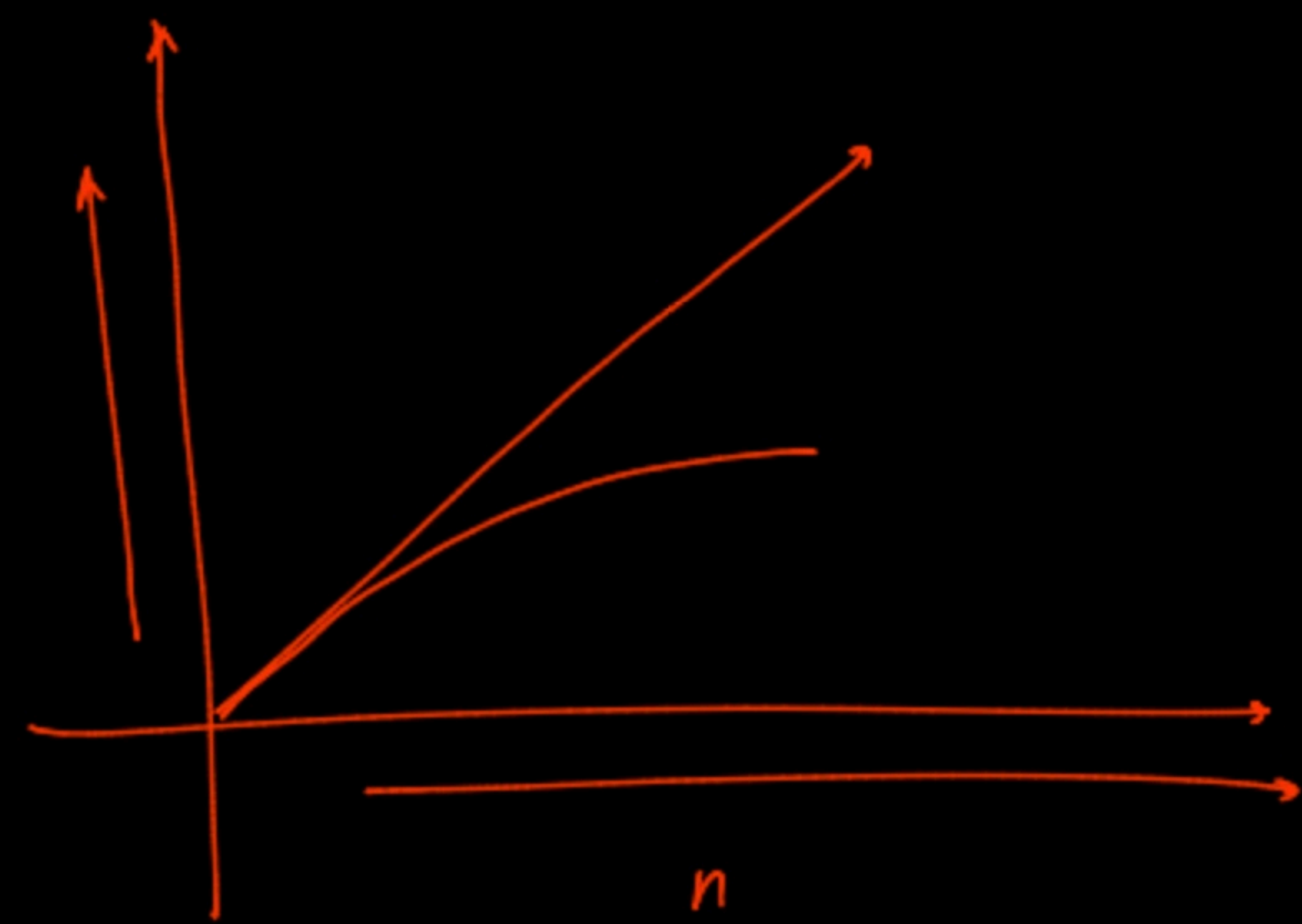
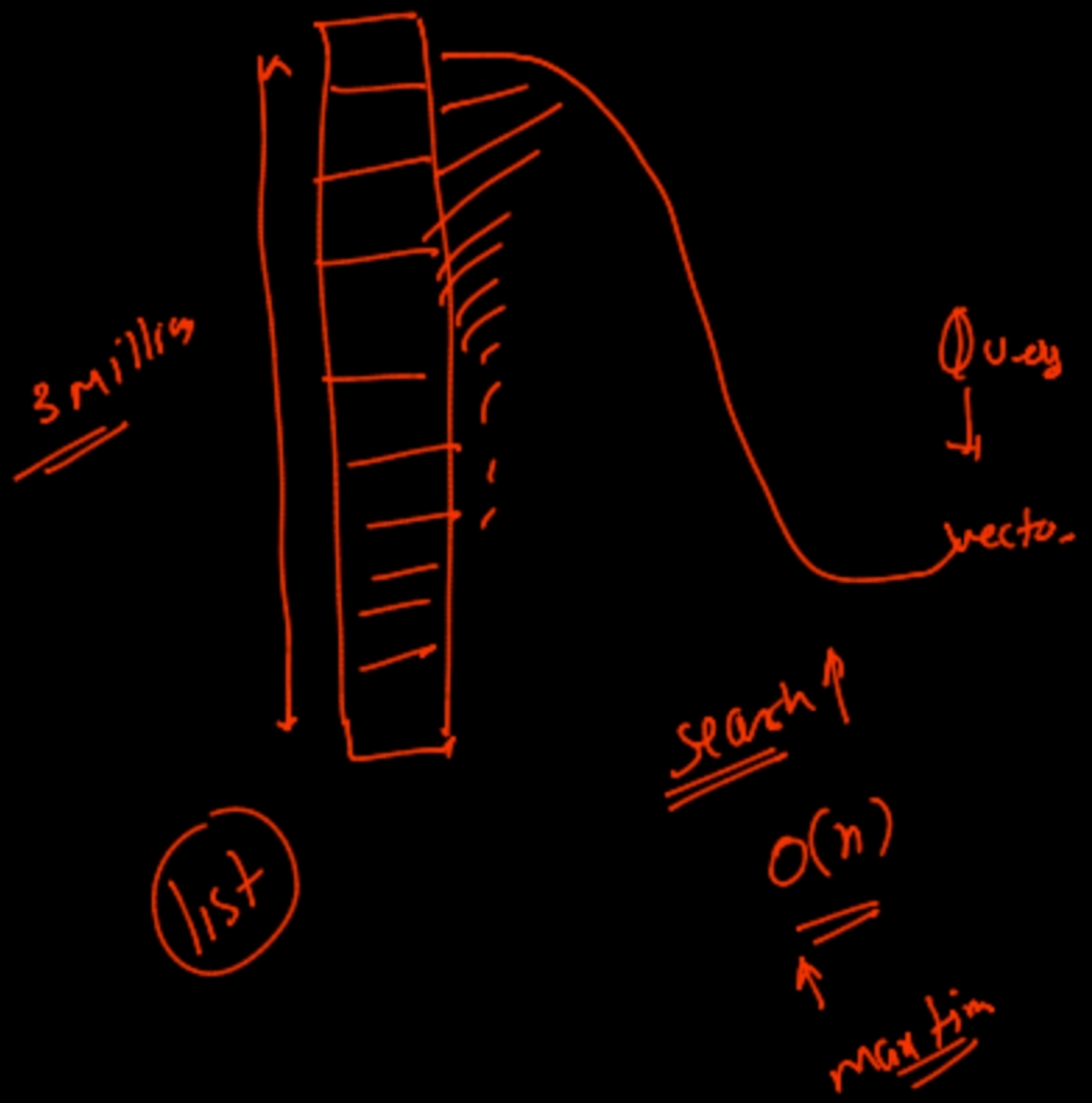
python code









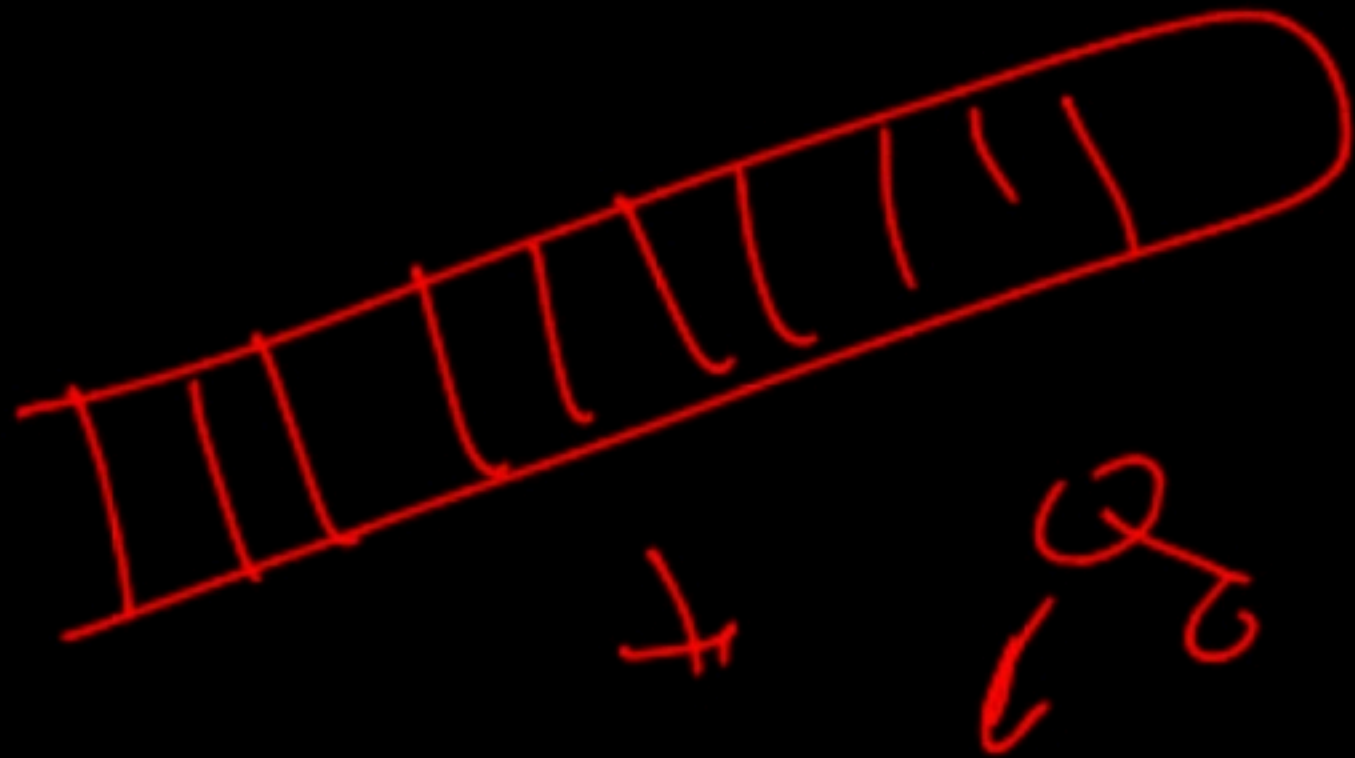


3m $\rightarrow$ 8-10 sec

||,

Data Struct

1-2 sec



$O(n)$ time

$O(\log n)$ time

$n = 8$

$\log 8$

$\log_2^3$

$3 * (\log 2)$

$(3) \rightarrow$

3000

time

$O(\log n)$ ✓

$O(\log 3000)$

$O(\log 2^{15})$

$15 \rightarrow 1642$

$(15) - 26$

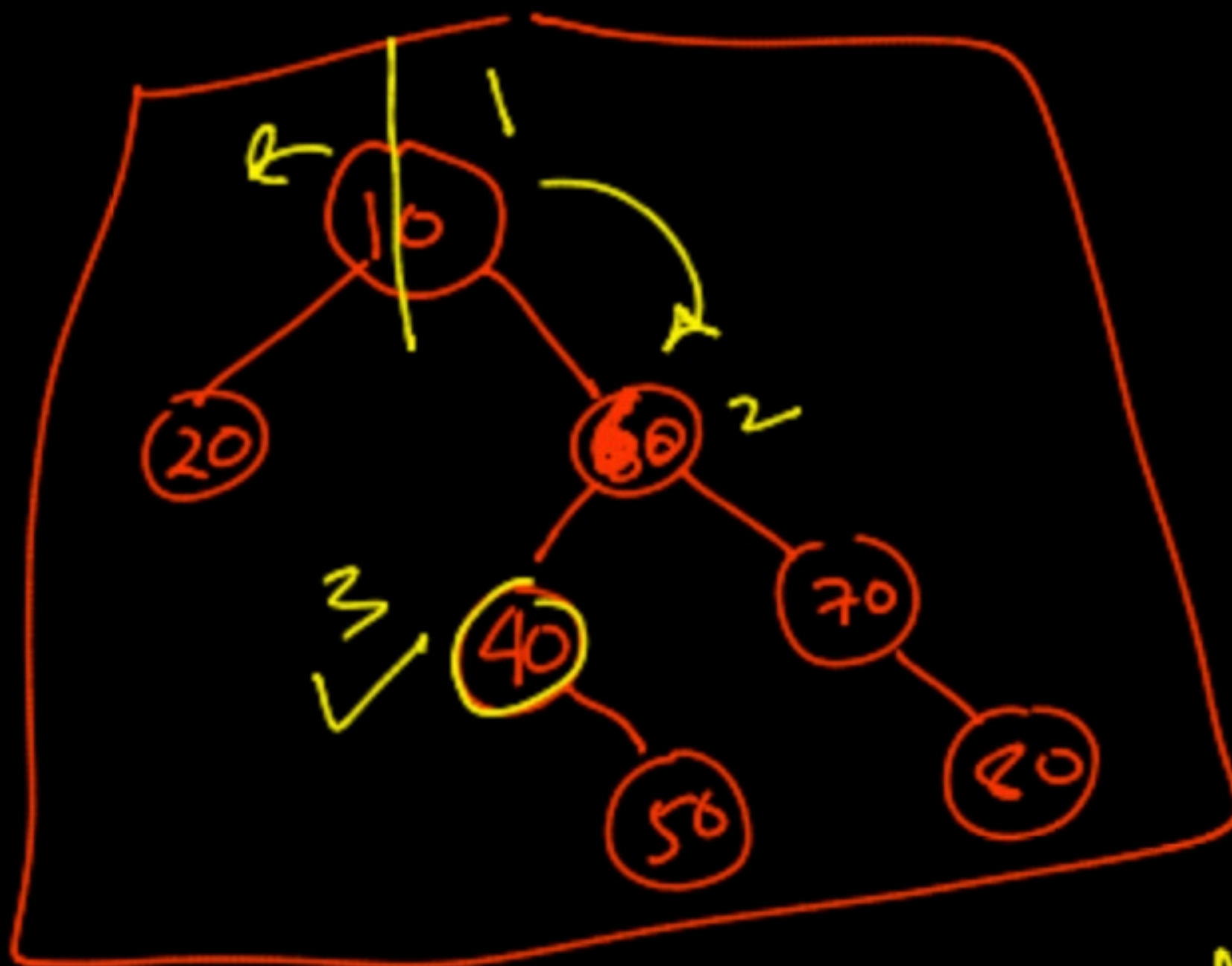
BST

[10, 20, 30, 40, 50, 60, 70, 80]

(i) insertion

tree

Worst $O(n)$



3 com

(ii) Search

40 $\geq, \leq$

SQL database
↓
B-Tree + Hashmap



$$V1 = [0.2, 0.3, 0.4, 0.5]$$

$$V2 = [0.1, 0.2, 0.3, 0.5]$$

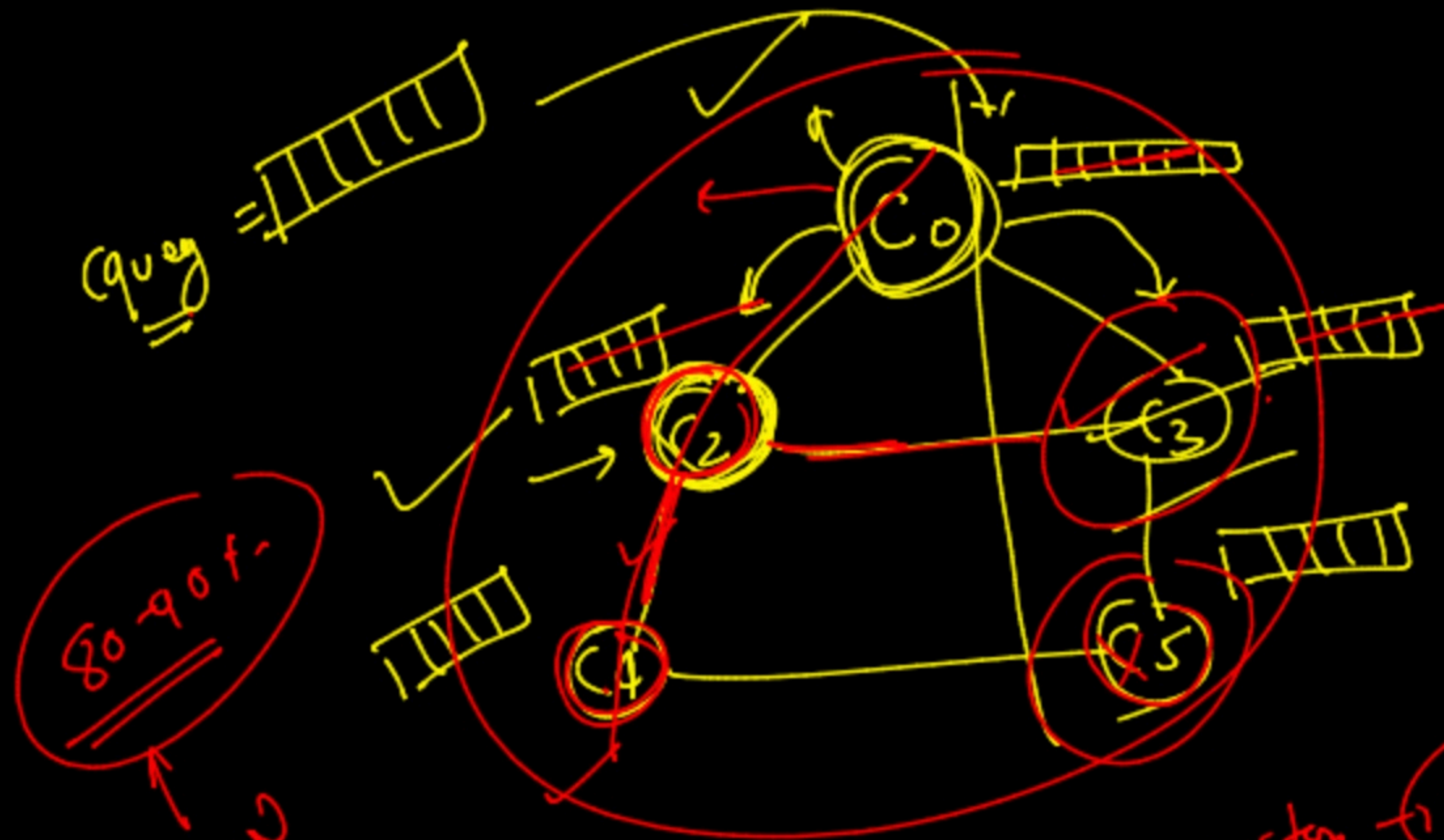
$$\left. \begin{array}{l} V1 > V2 \\ V2 < V1 \end{array} \right\} \times$$

I am good

I am not good

x





$$\begin{bmatrix} C_0 \\ C_2 \\ C_4 \end{bmatrix} \rightarrow \text{Output}$$

$O(n)$

$\downarrow$
 $O(\log n)$

$C_0, C_2, C_4, C_5 \rightarrow \text{vector} \rightarrow \text{graph}$

80-90%
APP

V1
V2
V3
!
!
!
V3max

PL



quij

